Full Pipeline Summary

What Goes In (Raw Inputs)

File	What It Is
`TOM_interactome_rep1/2/3.h5`	Ribosome profiling HDF5 files — TOM complex interactome, 3 replicates
`Translatome_rep1/2/3.h5`	Ribosome profiling HDF5 files — whole translatome, 3 replicates
`mitocarta.xls`	MitoCarta reference database of known mitochondrial proteins
`uniprot_proteome.tsv`	Full human UniProt proteome with sequences, gene names, and lengths reivewed

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What Comes Out (Final Output)

`all_genes_location_without_chx.csv` — one row per unique gene (16,170 total), with these key columns added:

Column	What It Contains
`gene`	Gene symbol (canonical where updated)
`protein_id`	NCBI protein accession
`cds_length`	CDS nucleotide length
`localization`	Final classification (see below)
`proteinLength`	Matched protein length in amino acids
`UniProt`	UniProt accession ID
`sequence`	Full protein amino acid sequence
`gene_old`	Original gene symbol before any canonical name updates

What Happens in Between

Notebook 1 (`all_genes.ipynb`) reads all 6 HDF5 files, deduplicates genes across replicates and conditions, and produces `all_genes_metadata_without_chx.csv` — a flat table of 16,170 unique genes with their `protein_id` and `cds_length`.

Notebook 2 (`localization.ipynb`) takes that CSV and runs it through a multi-stage classification and annotation pipeline:

```
16,170 genes
      │
      ├─ Match against MitoCarta by name + protein length
      │       → "mitochondrial" (confirmed), "name_only", "length_only",
"unclassified"
      │
      ├─ Cross-reference UniProt proteome by UniProt ID
      │       → upgrades unresolved genes to "mitochondrial" / "non-
mitochondrial"
      │
      ├─ Synonym rescue for MitoCarta entries missing from UniProt
      │       → catches genes known by alternate names
      │
      ├─ Secondary UniProt lookup by gene name + length
      │       → further resolves "name_only", "length_only", "unclassified"
      │
      ├─ NCBI Entrez batch fetch (for still-unresolved genes)
      │       → fetches protein sequences and GeneIDs from NCBI
      │
      ├─ GeneID → UniProt ID mapping via UniProt REST API
      │       → connects NCBI identifiers back to UniProt accessions
      │
      ├─ Fuzzy gene-name match (±1 aa tolerance)
      │       → last-resort catch for near-length matches
      │
      └─ Mark readthrough genes (gene name contains "-")
              → labeled "readthrough", excluded from mito/non-mito counts
```

Final Localization Labels

Label	Meaning
`mitochondrial`	Confirmed mitochondrial protein
`non-mitochondrial`	Confirmed cytoplasmic/other protein
`readthrough`	Read-through transcript (hyphenated gene name)
`name_only`	Name matches MitoCarta but length doesn't — likely an isoform
`length_only`	Length matches MitoCarta but name doesn't — ambiguous
`unclassified`	No match found in any reference source

In One Sentence

The pipeline takes raw ribosome profiling HDF5 data from 6 experiments, extracts every unique gene observed, and annotates each one with its subcellular localization (mitochondrial vs non-mitochondrial), its UniProt accession, and its protein sequence — using MitoCarta, UniProt, and NCBI as cascading reference sources.